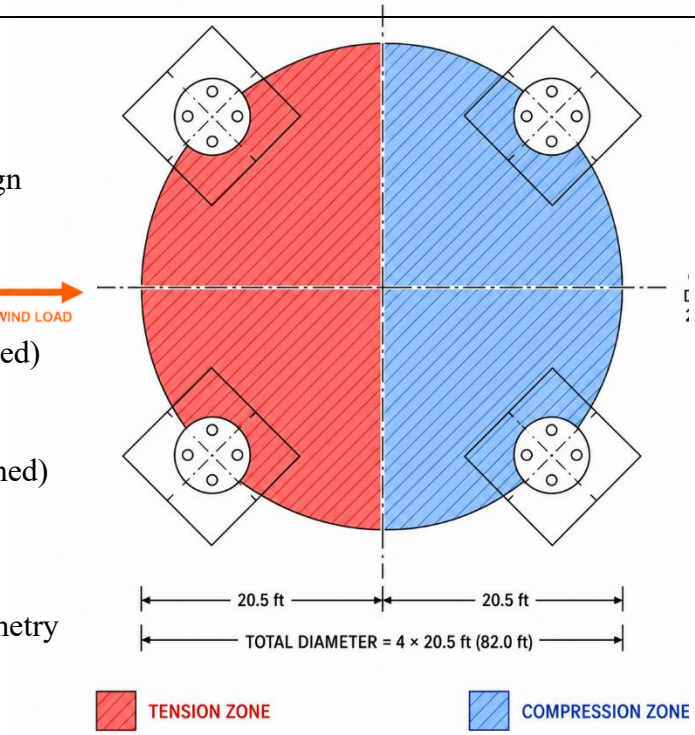
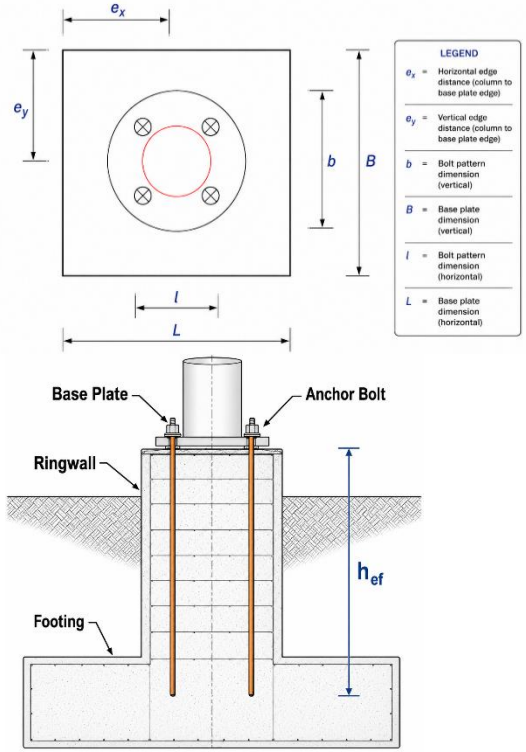
1. Anchor Bolts segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	1637.5732 kips	Main Tank Design (Predefined)
Factored moment (LRFD)	M_u	5673.437 kip-ft	Main Tank Design (Predefined)
Factored shear force (LRFD)	V_u	57.22 kips	Main Tank Design (Predefined)
Number of anchor bolts	N_b	4	User Input
Diameter circle representing axisymmetric spacing between legs/columns	D_{cl}	41′-0.5625″	Main Tank Geometry (Predefined)
Anchor bolt diameter	d_b	1.5 in	User Input
Effective embedment depth	h_{ef}	35 in	User Input
Concrete compressive strength	f'_c	4000 psi	User Input
Anchor bolt yield strength	F_y	36 ksi	User Input
Anchor bolt ultimate tensile strength	F_u	58 ksi	User Input
Edge distance	$e_x = e_y$	13.5 in	AISC Prescribed Minimum
Pedestal dimensions	B x L	39 in × 39 in	User Input
Bolt arrangement dimensions	b x l	12 in × 12 in	From the Pedestal Design
Strength reduction factors (tension, shear)	ϕ_t, ϕ_v	0.75, 0.75	ACI 318-19





STEP 1 — DETERMINE BOLT ARRANGEMENT DIMENSIONS (b x l)

e_x ..... Extracting from the table below.
e_y ..... Extracting from the table below.

Pedestal:

39"

Bolt spacing:

12"

Edge distance:

$$\frac{39 - 12}{2} = 13.5"$$

Required minimum:

$$7d = 7(1.5) = 10.5"$$

$b = B - 2 * e_x = 39-2(13.5)=12$ in
$l = L - 2 * e_y = 39-2(13.5)=12$ in

STEP 2 — DETERMINE TENSION SIDE LEGS

For overturning, assume:

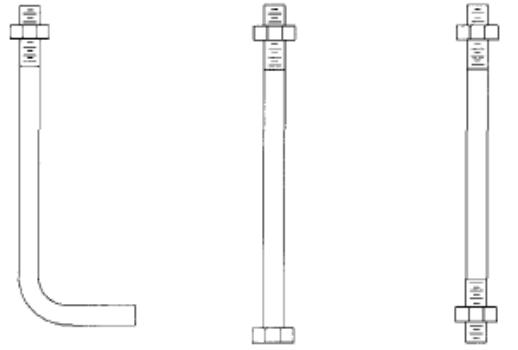
- 2 legs in compression
- 2 legs in tension

Thus…….**nt=2**

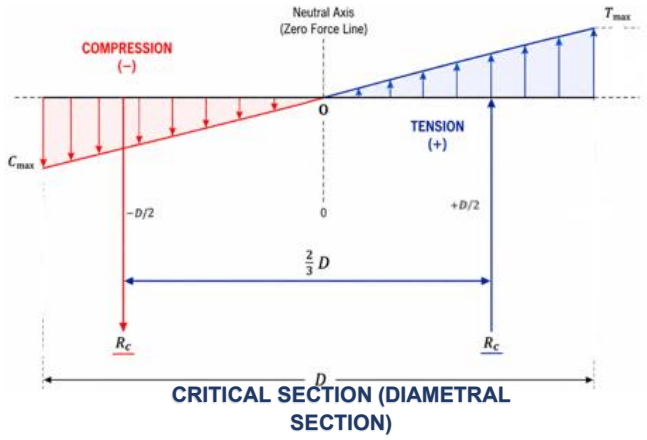
Minimum Bolt Lengths and Edge Distances

Shipp and Haninger (1983) have presented minimum guidelines for bolt embedment and edge distance, adopted from ACI 349. These are presented for use in designing anchor bolts for tension as follows:

Bolt Type, Material	Minimum Embedded Length	Minimum Embedded Edge Distance
A307, A36	12 *d*	5 *d* > 4 in.
A325, A449	17 *d*	7 *d* > 4 in.



(a) Hooked Bar *(b) Bolt* *(c) Threaded Bar with Nut*



CRITICAL SECTION (DIAMETRAL SECTION)

STEP 3 — DETERMINE LEG RADIUS

Given……

$$D = 41' - 0\frac{9}{16}"$$

Radius:

$$r = 20.523'$$

Convert:

$$r = 246.28"$$

STEP 4 — TENSION FORCE FROM OVERTURNING

The overturning moment obtained from the critical section (Step 2) is used to determine the tensile force developed in the anchor-bolt group.

$$T = \frac{M_u}{4/3r}$$

Convert moment:

$$M_u = 5673.437 \times 12$$

$$M_u = 68081.24 \text{ kip-in}$$

Now:

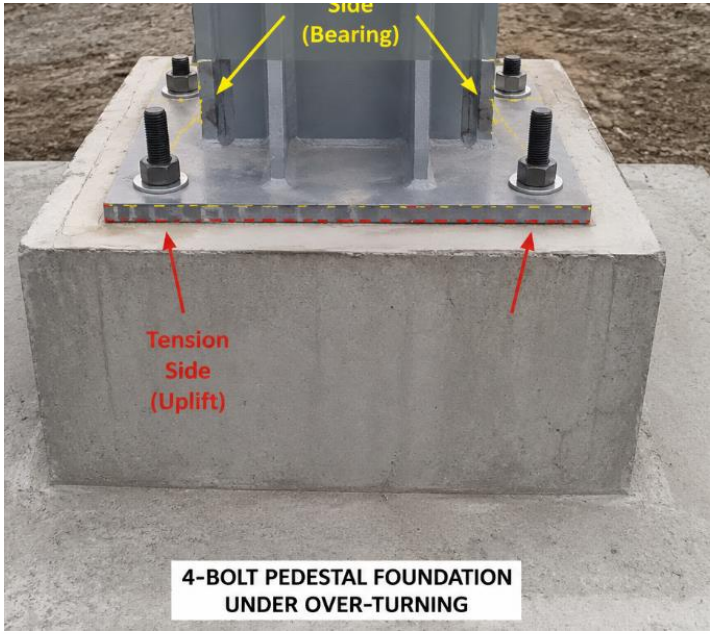
$$T = \frac{68081.3}{\frac{4}{3} * 246.28}$$

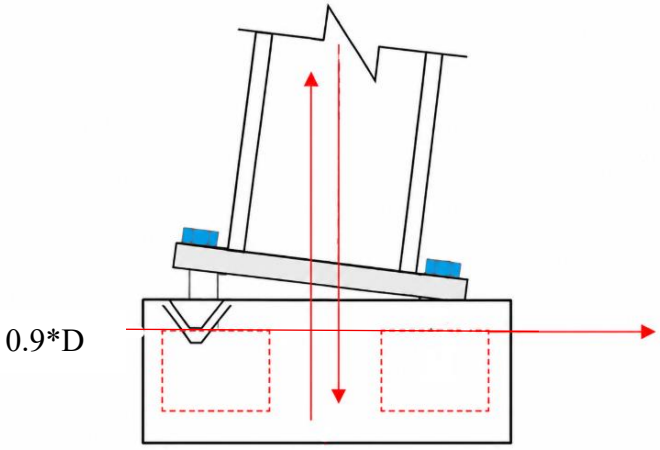
$$T = 207.329\ kips$$

This is total overturning uplift.

Shared by 2 tension legs:/

$$T_{leg} = \frac{207.329}{2}$$





Anchor bolt failure
(tension side)

$$T_{leg} = 103.6644\ kips$$

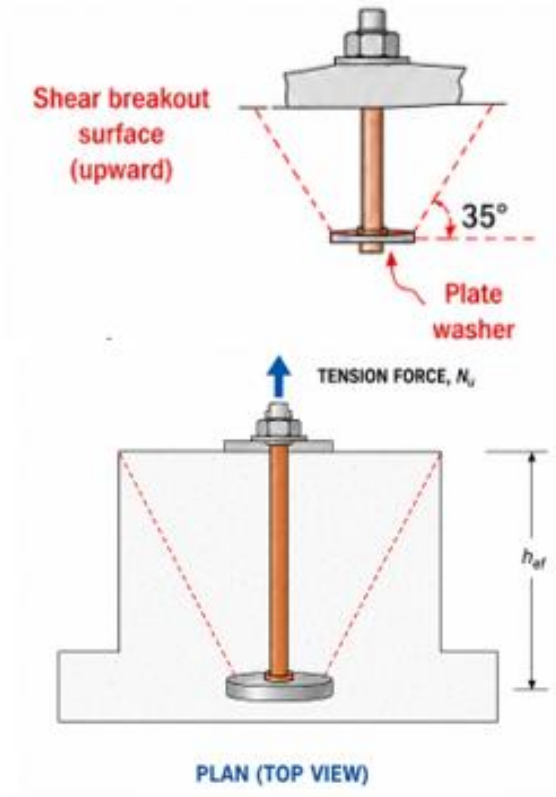
Note

- The calculated force kips is the tensile force used for the anchor bolt tension design.
- The compression side resists the remaining overturning force through bearing between the base plate and the concrete foundation.
- The overturning moment has already been determined at the critical section in Step 2; therefore, this step only converts that moment into the corresponding anchor-bolt tensile force.

STEP 5 — TENSION PER BOLT

$$T_{bolt} = \frac{103.6644}{4}$$

$$T_{bolt} = 25.9161\ kips$$



STEP 6 — REQUIRED STEEL AREA

LRFD:

$$\phi = 0.75$$

Required steel area:

$$A_s = \frac{T_u}{\phi F_y}$$

$$A_s = \frac{25.9161}{0.75(36)}$$

$$A_s = 0.96\ \text{in}^2$$

STEP 7 — SELECT BOLT DIAMETER

Diameter	Area
1.5 in	1.77 in²
1.75 in	1.90 in²
2.0 in	2.50 in²
2.25 in	3.25 in²

Based on this we can select:

$$A_s = 1.5997\ in^2$$

Bolt Type, Material	Minimum Embedded Length	Minimum Embedded Edge Distance
A307, A36	12 *d*	5 *d* > 4 in.
A325, A449	17 *d*	7 *d* > 4 in.

SELECT

1.5" bolt is sufficient

Area:

$$A_b = 1.77\ \text{in}^2$$

Table 17.5.3(a)—Anchor strength governed by steel

Type of steel element	Strength reduction factor ϕ	
	Tension (steel)	Shear (steel)
Ductile	0.75	0.65
Brittle	0.65	0.60

STEP8 — CHECK STEEL SHEAR CAPACITY

Shear per pedestal:

$$V_{bolt} = \frac{V_u}{N_b\ x\ N_{leg}}$$

$$V_{bolt} = \frac{57.218}{4x4}$$

$$V_{bolt} = 3.576\ \text{kips}$$

Steel shear capacity:

$$\phi V_s = 0.75(0.6)(36)(1.50)$$

$$\phi V_s = 24.3\ \text{kips}$$

Check:

$24.3 > 3.576$……..**PASS**

STEEL TENSION CAPACITY

$$\phi N_s = \phi A_b F_y$$

$$\phi N_s = 0.75(1.77)(36)$$

$$\phi N_s = 47.8\text{kips}$$

STEP 9 — CONCRETE BREAKOUT CHECK

ACI approximate breakout: (17.6.2 and 17.3.3).

Add a note adjacent to the input box indicating the required information. Also,

display the corresponding Table 17.5.3(a) near the input box for user reference.

$$N_{cb} = \phi k_c \sqrt{f_c'} h_{ef}^{1.5}$$

Use:

- $k_c = 24$
- $\phi = 0.75$

Compute:

$$\sqrt{4000} = 63.25$$

$$35^{1.5} = 207.1$$

$$N_{cb} = 0.75(24)(63.25)(207.1)$$

$$N_{cb} = 235783.4 \text{ lb}$$

$$N_{cb} = 235.7 \text{ kips}$$

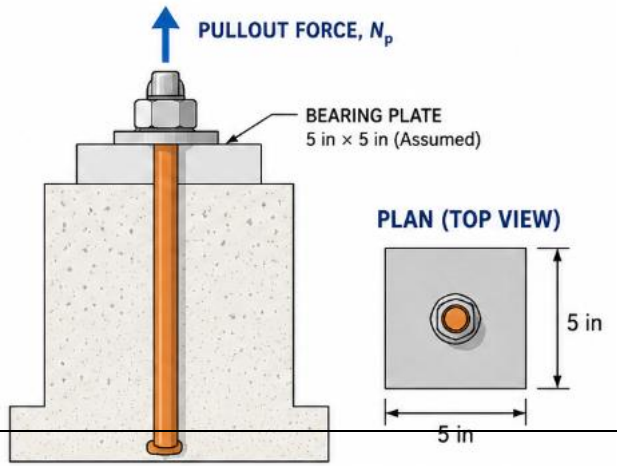
Check:

$$235.8 > 25.91 \ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

STEP 10 — PULLOUT CHECK

Pullout strength shall be verified using the actual anchor head or bearing plate dimensions provided in the final anchor detail. For preliminary design, a 5 in × 5 in bearing plate was assumed.

$$A_{brg} = 25 \text{ in}^2$$



Pullout:

$$N_p = 8A_{brg}f_c'$$

A_{brg}= bearing area of anchor head or plate washer (Assume a 5 in × 5 in square plate washer)……..

ACI 318 Chapter 17 requires pullout strength to be based on the actual anchor geometry

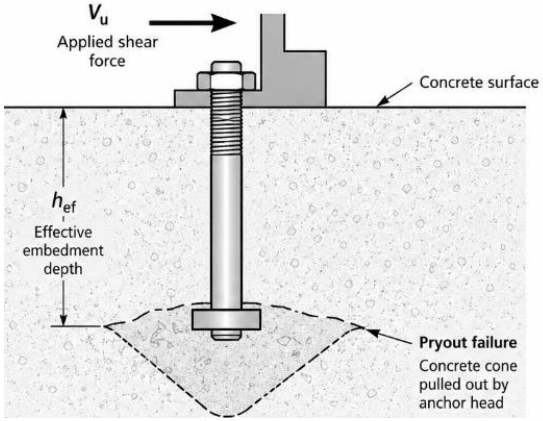
$$N_p = 8(25)(4000)$$

$$N_p = 800000 \text{ lb}$$

$$N_p = 800 \text{ kips}$$

Check:

$$800 > 24.3 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$



STEP 11 — PRYOUT CHECK

ACI 318-19 Chapter 17 (Anchoring to Concrete)

$$V_{cp} = k_{cp} * N_{cb}$$

where:

- V_{cp}= pryout strength
- N_{cb}= concrete breakout strength in tension
- k_{cp}= pryout factor

$$V_{cp} = 2.0 N_{cb}$$

$$V_{cp} = 576 \text{ kips}$$

k_{cp} = 1.0 for h_{ef} < 2.5 in.
k_{cp} = 2.0 for h_{ef} ≥ 2.5 in.

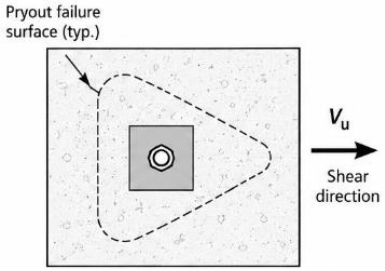
ACI 318-19 Table 17.7.3.1 (Pryout Strength Factor k_{cp})

Check:

$$576 > 3.576 kips \ldots\ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

STEP 12 — INTERACTION CHECK

Based on Section 17.8.3. ACI 318-19,



$$\left(\frac{N_u}{\phi N_n}\right)^{5/3} + \left(\frac{V_u}{\phi V_n}\right)^{5/3} \leq 1.0$$

Compute:

$$\left(\frac{34.55}{47.8}\right)^{5/3} + \left(\frac{3.576}{24.3}\right)^{5/3}$$

$$= (0.7228)^{5/3} + (0.163512)^{5/3}$$

$$= 0.5664$$

Check: $0.5664 < 1.0$…… **PASS**

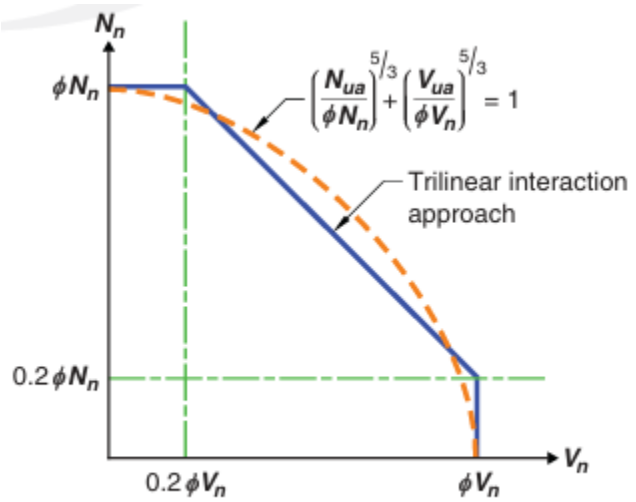


Fig. R17.8—Shear and tensile load interaction equation.